Light and Atomic Spectra

Answer key with worked solutions

A. Waves and photons

1. Order of energy.

> From lowest to highest energy: **radio waves → microwaves → infrared → visible → ultraviolet → X-rays**

2. Microwaves.

> (a) 12.2 cm = 0.122 m
>
> $$\nu = \frac{c}{\lambda} = \frac{2.998 \times 10^{8}\ \text{m/s}}{0.122\ \text{m}} = \mathbf{2.46 \times 10^{9}\ s^{-1}}$$
>
> (b)
>
> $$E = h\nu = 6.626 \times 10^{-34}\ \text{J}\cancel{s} \times 2.457 \times 10^{9}\ \cancel{s^{-1}}$$
>
> $$= \mathbf{1.63 \times 10^{-24}\ J}$$

3. Blue light.

> $$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m/s}}{450 \times 10^{-9}\ \text{m}} = \mathbf{4.41 \times 10^{-19}\ J}$$
>
> Per mole: $4.414 \times 10^{-19}\ \text{J} \times 6.022 \times 10^{23}\ \text{mol}^{-1} = 2.66 \times 10^{5}\ \text{J/mol} = \mathbf{266\ kJ/mol}$

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4. UV and bond breaking.

(a)

$$E = \frac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m/s}}{254 \times 10^{-9}\ \text{m}}$$
$$= \mathbf{7.82 \times 10^{-19}\ J}$$

(b) Energy per bond:

$$E = \frac{498 \times 10^{3}\ \text{J/mol}}{6.022 \times 10^{23}\ \text{mol}^{-1}} = 8.270 \times 10^{-19}\ \text{J}$$
$$\lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m/s}}{8.270 \times 10^{-19}\ \text{J}}$$
$$= 2.40 \times 10^{-7}\ \text{m} = \mathbf{240\ nm}$$

Longer wavelengths carry less energy per photon, so 240 nm is the longest that works. (The 254 nm lamp is just too weak to break O=O.)

B. Line spectra and the Bohr model

5. Spectra.

A **continuous spectrum** contains all wavelengths (a hot solid). A **line spectrum** contains only certain exact wavelengths (an excited gas). Each line is a photon with energy equal to the gap between two levels; since only certain wavelengths appear, only certain energies are allowed: the levels are **quantized**.

6. Bohr calculations.

(a)

$$\Delta E = 2.179 \times 10^{-18}\ \text{J}\left(\tfrac{1}{2^2} - \tfrac{1}{4^2}\right) = 4.086 \times 10^{-19}\ \text{J}$$
$$\lambda = \frac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m/s}}{4.086 \times 10^{-19}\ \text{J}}$$
$$= 4.86 \times 10^{-7}\ \text{m} = \mathbf{486\ nm}\ (\text{visible, blue-green})$$

(b)

$$\Delta E = 2.179 \times 10^{-18}\ \text{J}\left(\tfrac{1}{1^2} - \tfrac{1}{2^2}\right) = 1.634 \times 10^{-18}\ \text{J}$$
$$\lambda = \frac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m/s}}{1.634 \times 10^{-18}\ \text{J}}$$
$$= 1.22 \times 10^{-7}\ \text{m} = \mathbf{122\ nm}\ (\text{ultraviolet})$$

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7. Ionization from n = 2.

> $E_2 = -\dfrac{2.179 \times 10^{-18}\ \text{J}}{2^2} = -5.448 \times 10^{-19}\ \text{J}$, so the energy needed is $\mathbf{5.45 \times 10^{-19}\ J}$ per atom.
>
> Per mole: $5.448 \times 10^{-19}\ \text{J} \times 6.022 \times 10^{23}\ \text{mol}^{-1} = 3.28 \times 10^{5}\ \text{J/mol} = \mathbf{328\ kJ/mol}$

C. Challenge

8. One photon or one mole?

> 182 kJ is the energy of a **mole** of photons (6.022×10^{23} photons), not of one photon. For one photon:
>
> $$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 2.998 \times 10^{8}\ \text{m/s}}{656 \times 10^{-9}\ \text{m}}$$
> $$= \mathbf{3.03 \times 10^{-19}\ J}$$

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